Fiscal Year:	FY 2015	**Task Last Updated:**	FY 02/19/2015
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Microgravity and Magnetic Fields - Directional Root Growth and Gene Expression		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/03/2010	**End Date:**	10/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: New end date is 10/31/2015 per NSSC information (Ed., 3/3/15) NOTE: New end date is 3/31/2015 per NSSC information (Ed., 1/22/14) NOTE: New end date is 12/31/13 per NSSC information (Ed., 8/6/13)		
Key Personnel Changes/Previous PI:	n/a		
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Task Description:	This proposal combines morphological measurements on micro-gravity germinated and grown (Brassica) roots that are exposed to high-gradient magnetic fields (HGMF) with a high-resolution spatial analysis of genes involved in gravitropism. Magnetic gradients exert a repulsive force on diamagnetic substances such as the starch in amyloplasts, which results in gravitropic-like curvature. The proposal is based on hardware that was developed for a previous flight experiment on STS-107. As in the originally proposed experiment, microscopic analysis of amyloplast size and distribution will be studied in the context of achieved curvature. In addition, high-resolution genetic profiling will be conducted that focuses on the cytoskeleton and starch development. The result of the proposed work will enable directional growth of seedlings and eventually mature plants under microgravity conditions using HGMF. Three conditions will be used to differentiate microgravity from uniform magnetic fields and HGMF conditions. Only HGMF is expected to influence root curvature. A uniform magnetic field will serve as control for possible secondary magnetic effects. This condition will be compared to an in-flight control without magnetic field to determine growth rate, gene expression, growth, and directionality (autotropism). Root material from these experiments will be used for investigations on cytoskeletal organization and the expression of starch-related genes. A second aspect of the proposed work will investigate the process of gravisensing, whether dependent on amyloplast position, movement, or force exerted onto amyloplasts. Space experimentation is necessary to establish a threshold value for a directional magnetic force. In addition to providing insight in the fundamental mechanism of the gravisensing system of plants, the data from this experiment can be used to develop technology to provide a directional, gravity-like stimulus to plants during early stages of germination. The proposed experiment can be performed with existing hardware, be adapted easily to changing flight schedules, and requires minimal crew time. The anticipated benefits include a better understanding of root development, effects of amyloplast displacement on root development and curvature, and compensatory measures to alleviate microgravity conditions. A newly developed gene extraction technology is ideally suited for gene-profiling studies and has far-reaching implications for studying gene expression, especially for confined space flight experiments. Additional benefits of the project include training of students at the undergraduate and graduate levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to an improved understanding of basic biological questions, such as identification of the site of graviperception, curvature-induced (but not gravity-dependent) gene expression, the experiment also tests the effect of magnetic fields on plant roots. Therefore, biological questions are investigated in conjunction with physical principles. We intend to analyze the effect of static magnetic fields on biological systems. This work will provide a starting point for the current discussion of the detrimental effects of (electro)magnetic fields. In addition, understanding basic biological questions will improve the prospect for enhanced plant productivity, resource optimization, and reduced maintenance.
	Research activities during the last 16 months included continued research on brassica growth responses, procedural modifications for the planned flight, launch preparations, launch of the flight experiment on Space-X 3, preliminary evaluation of the flight data, and two attempts of a clinorotated ground control. Continued research on brassica biology and growth response included the genetic analyses of the effect of reduced oxygen on seedling metabolism. The rationale behind this study concerns two aspects: the determination of the maximal number of seeds that can grow in a confined volume and the response to space conditions where the re-supply of gases occurs in the absence of convection or where convection is reduced and dependent on artificial ventilation. In addition, oxygen consumption for most seeds is highest during germination and the absorbing surface is minimal. The combination of these effects may lead to O2 deprivation. The tests were conducted with seeds grown in 0, 2.5, 5, 10, and 21% oxygen. Nine genes were tested. Oxygen concentrations below 10% reduced growth and below 5% delayed germination. In the absence of oxygen germination did not occur; however, re-introduction to air resulted in germination in less than 6 hours. These seeds reach the same growth rate as air-grown seedlings within 12 hours. Hypoxia inhibited most genes, especially Amylase (AMY), and sucrose synthase (SuSy), fructose 1,6-P aldolase (ALD), but up-regulated Glucose-P isomerase (GPI). These data have been reported at the Committee for Space Research (COSPAR) meetings in Moscow, Russia and are the basis for a future publication. Procedural testing and modifications for the planned flight included new braces for the magnetic field chamber (MFC) covers, gaskets for the fixative storage container, and changes in the default fixation time. Preparations also included image modification of the imaging board to minimize distortions. These efforts were based on test images of 1 mm grids and resulted in the development of a software (Photoshop) script. Launch preparations started 12/2013 but various delays (including 4 unsuccessful launch attempts) delayed the launch until April 18, 2014, 3:25 pm EDT. The experiment was transferred to the International Space Station (ISS) express rack 10 days later and the experiment started May 7 and fixative was dispensed 47 hours and 25 minutes later, and terminated after successful fixation at 48 h and 30 minutes. Transmitted imaged of growing seedlings showed overall successful germination and, most significantly, root curvature away from the high-gradient magnetic field (HGMF). The experiment was transferred back to the Dragon capsule (5/15/2014@13:55), Dragon undocked (5/18/2014@9:45), and landed successfully (5/18/2014@15:12). The experiment arrived at the Kennedy Space Center (KSC) (5/22/2014@1:05) and de-integrated within 6 hours. Removal of the seeds cassettes from the MFC's showed substantial effects of RNALater storage (total time: 12 days and 22 hours). Corrosion of the NdFeB magnets and presence of high concentrations of ammonium sulfate (main ingredient of RNALater) produced FeSO4 and released ammonium gas. This gas pressurized the MFC and forced most of the fixative into the waste bag that eventually burst and released significant quantities of fixative into the generic containment unit. The waste bag in the non-magnetic MFC remained intact and the MFC with formaldehyde fixative did not produce corrosion to the extent seen for RNALater. Despite these unexpected effects, fixation was successful and the tissue could be processed for gene expression analysis and confocal microscopy. Gene expression was determined in the root tip (~3 mm), remainder of the root, root/shoot junction, and the seed. The selection is based on recent discoveries of a second gravitropic response that involved the root/shoot junction. This phenomenon is the topic of a dissertation by one of the PI's graduate students. Sixteen genes were selected that are related to previous studies and contained reference genes (UBQ1, COX), cytoskeletal and transport related genes (ACT7, TUB1), auxin response genes (IAA5, PIN3, PIN7), genes related to carbohydrate metabolism (ADH1, AMY, G6PDH5, GLK, HXK, PFK, SUSY), stress metabolism (ADH1), and lipid metabolism (TAGL). The preliminary evaluation was based on the comparison of flight data ± experimental condition (e.g., magnetic vs. non-magnetic MFC; flight, static ground control etc.). These conditions prompted research into the determination of up- or down-regulated genes in the various tissues and in consultation with a colleague at the Louisiana Accelerator Center,

Task Progress:	Dr. Louis Houston, an outlier theorem was developed that allows the identification of significant changes in gene expression. This identification utilizes non-redundant data sets and identifies z = 2‹xi› as the outlier boundary. 2‹xi› = N+2z0/(N+1). Since N+z0/(N+1) is continuous on N, 2‹xi› is the upper outlier boundary for all values of N. Based on this result, gene activation can be derived as the calculated unique distance of the mean between two data sets. This description identifies significant upregulation of a subset of genes in the presence of HGMF. Upregulated genes include: ADH1, AMY, GLK, HXK, IAA5, SuSy, PIN1, PIN3, and PIN7. Additional evaluation concerns the identification of effects between HGMF in weightlessness and 1-g conditions and clinorotation. Interestingly, clinorotation upregulates amylase (starch degrading enzyme). Microscopic analyses focus on two aspects, distribution of amyloplasts and organization of cytoskeletal proteins. Unexpectedly, the space flight material showed that columella amyloplasts are ~15% larger than ground controls. This observation indicates that weightlessness triggers more resources to go into gravisensing structures. Future experiments should investigate if the g-sensing threshold in space might be lower than the reference values (ca. 10-3 g) that were obtained from clinorotation experiments (Shen-Miller et al., 1968), which enhances starch degradation. The examination of the cytoskeletal organization revealed less parallel microtubules (MTs) and thinner actin filaments (AFs) in space-grown material than in static ground controls. A forthcoming analysis will be based on studies of the alignment and consistency of MTs and AFs based on an ImageJ plug-in (Orientation-J) that computationally determines coherency (parallelism) and orientation of cytoskeletal proteins. The first ground control was attempted August 13, 2014. Unfortunately, the compromised MFC (see flight experiment described above) released toxic (ammonia) fumes upon imbibition and the seeds failed to germinate. In addition to the lack of germination, the clinorotation negatively affected the fixation system and RNALater leaked into the generic containment unit (GCU). Luckily, none of the electronics were affected and the ground control could be repeated with MFCs that were not exposed to RNALater. Because of the failure to achieve germination, tests were done at the PI's lab that revealed additional, apparently clinorotation-specific problems. Although the seeds germinated, the apical hook formation was excessive and prevented the majority of the roots to extent toward the magnetized wedges. This condition was remedied by placing the seeds at the center of the germination paper in the seed cassettes. While this measure was a deviation from the original protocol, it provided data that allowed for a comparison between clinorotated and flight data. The advantage of the clinorotation of the original flight hardware was the imaging capability that is not available in the PI's lab. Brassica rapa experiments were replicated twice in a second MFC using the following conditions: A) Vertical, no magnets, no clinorotation ; B) Vertical, with HGMF, no clinorotation ; C) Clinorotation, no magnets (comparison to KSC experiment – MFC3) ; D) Clinorotation, with HGMF (comparison to KSC experiment – MFC2). The gene analyses from these experiments represent a comparison to the analyses obtained from the second ground control performed at KSC 1/7 to 1/9/2015. These data are still being analyzed but both germination and elongation-growth were nominal and therefore successful. MFC2 contained the critical combination of HGMF and RNALater. Although significantly reduced in length, germination was not affected. Current analyses are underway to tease out which processes might be affected by ammonia exposure. References Shen-Miller J, Hinchman R, Gordon SA (1968) Thresholds for Georesponse to acceleration in gravity-compensated avena seedlings. Plant Physiology 43: 338-344
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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